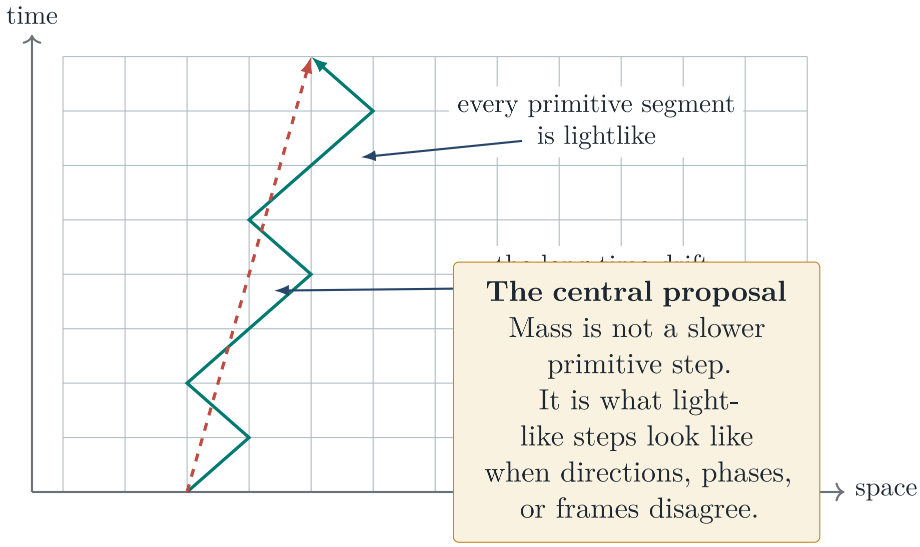
Reality in Null Steps

A visual guide to mass, motion, and geometry on a finite causal grid

Null-Edge Formalization Project

General-audience companion manuscript, July 9, 2026



This manuscript explains a finite mathematical research program. It does not claim that experiments have established a literal spacetime grid, that an electron follows a hidden classical zigzag, or that measured particle masses have been derived. Bold pictures are paired with their precise boundaries.

Start from something you already believe

Most of the mass of ordinary matter is already known not to come from massive building blocks. ESTABLISHED PHYSICS About 99 percent of the mass of every proton and neutron — and therefore most of your own weight — is the energy of nearly massless quarks and exactly massless gluons confined in a small volume. This is standard quantum chromodynamics, memorably summarized by Frank Wilczek as "mass without mass" [2, 3]. In the same spirit, a mirrored box filled with light weighs more than the same box empty: two flashes of light moving in opposite directions have no individual mass, yet together they have a rest frame and a positive invariant mass.

So the question explored here is not whether mass can emerge from massless ingredients — for most of the matter around you, it demonstrably does. The question is whether the pattern goes all the way down: whether *every* elementary process is lightlike, and every mass, without exception, is the cost of organizing lightlike processes into one stable object. That stronger claim is not established physics. It is a research program, and this manuscript explains both the picture and its current limits.

In brief

Suppose the most elementary propagation allowed by nature is always lightlike: one step forward in time comes with one null step through space. A massive object can nevertheless drift slower than light if its quantum amplitude is shared among different null directions. In this picture, mass measures the failure of many primitive directions to behave as one perfectly aligned ray.

We develop that idea on a finite causal grid. Four recurring structures appear when a Dirac-like transport operator is squared: *aperture*, the opening between directions; *closure*, the mismatch accumulated around a loop; *turn*, the conversion between chiral propagation channels; and *soldering*, the map that ties internal transport to spacetime geometry. Several finite identities behind this picture have been checked by the Lean proof assistant, a computer program that verifies every logical step of a proof (Section 9). The extension from those finite models to the observed Standard Model and continuum spacetime remains a falsifiable research program.

How to read the claims

Three kinds of statement are deliberately kept separate.

Picture	A way to see the mechanism: null arrows, loops, turns, and frames.
checked finite result	An exact finite statement accepted by Lean's kernel for the displayed model.
established physics	Standard mathematics or physics imported from established literature.
open interpretation	A proposed identification with continuum fields, particles, or measured reality.

The same sentence can be useful at one level and false at another. "A massive particle is made from lightlike motion" is exact as a momentum decomposition, suggestive as a quantum-walk picture, and unproved as a literal microscopic ontology. The colored tags introduced above therefore appear throughout the text, marking claim by claim which tier is active.

1 The picture in one page

Imagine a clock that advances in indivisible ticks. At each tick, an elementary amplitude may propagate only along one of a finite set of null links. No link is timelike. Nothing elementary waits in place.[1]

Now allow the amplitude to use more than one null direction (Figure 1). A sequence can zig right, zag left, change an internal phase, or be transported around a loop. The individual links remain null, but their combined momentum need not be. The net object can possess a rest frame and a positive invariant mass.[2]

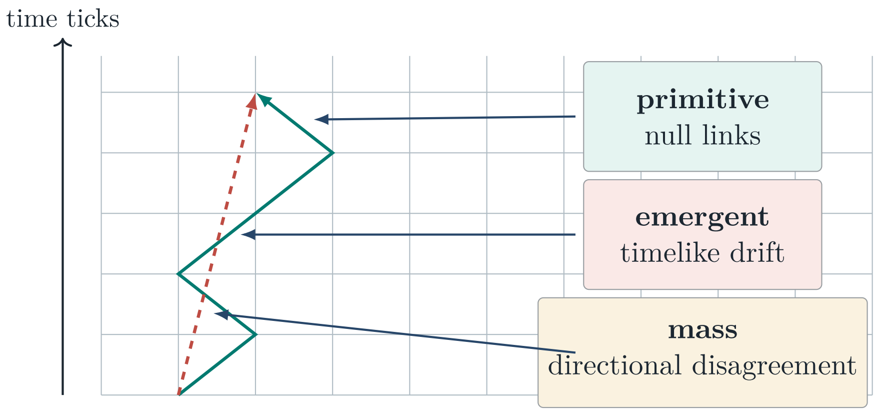


Figure 1: Null microscopic support and timelike coarse drift are different statements. The dashed line is an average direction, not an additional microscopic link.

Four questions then organize the model:

APERTURE
Do the directions open apart?

CLOSURE
Does a loop return unchanged?

TURN
Does propagation switch channel?

SOLDER
Do internal and spacetime frames agree?

These are not four metaphors attached after the fact. They are the four structural kinds of term that appear in the square of the finite *carrier* — the single operator, studied throughout the project, that advances the quantum amplitude by one tick. OPEN INTERPRETATION Whether the four kinds become kinetic, gauge, Yukawa, and gravitational terms in a continuum theory is the central open test.

[1] This is a statement about the support of the model's one-step propagator. In units with $c = 1$, each allowed displacement $\ell = (\Delta t, \Delta \mathbf{x})$ satisfies $\ell^2 = \Delta t^2 - |\Delta \mathbf{x}|^2 = 0$. It is not yet a claim that physical spacetime has a smallest experimentally observable tick.

[2] The rigorous finite statement is made with the positive Hermitian momentum matrix $P = \sum_i \psi_i \psi_i^\dagger$. Each summand has determinant zero, while $\det P$ can be positive. The language of a single object following the drawn zigzag is interpretive; quantum amplitudes are summed over histories.

2 What "everything moves at the speed of light" can mean

The slogan is memorable, but it needs three distinctions.

Primitive speed

Each allowed microscopic link lies on the light cone. In units where $c = 1$, a step has the form

$$(\Delta t, \Delta \mathbf{x}) = (1, \mathbf{n}), \qquad |\mathbf{n}| = 1,$$

so its interval is $\Delta t^2 - |\Delta \mathbf{x}|^2 = 0$.

Front speed

ESTABLISHED PHYSICS For a relativistic wave equation, the fastest disturbance is governed by the highest-derivative, or *principal*, part of the equation. If that part is

$$\omega^2 - |\mathbf{k}|^2,$$

the wavefront lies on the null cone. Finite mass matrices and other lower-order internal couplings do not change this front cone. They do change dispersion and interference behind the front.[3]

Group drift

ESTABLISHED PHYSICS A massive wave packet obeys

$$E^2 = |\mathbf{p}|^2 + m^2, \qquad |\mathbf{v}_{\text{group}}| = \frac{|\mathbf{p}|}{E} < 1 \quad (m \neq 0).$$

There is no contradiction. The support of the microscopic propagation and the drift of a massive packet answer different questions.

The strongest honest slogan. CHECKED FINITE RESULT All primitive propagation in the model is null; massive motion is an emergent timelike drift produced by coherent mixing among null channels.

ESTABLISHED PHYSICS For a Dirac fermion, each individual velocity component has spectral values $\pm c$. But the three component operators do not commute, so they do not define a simultaneous classical three-velocity of fixed magnitude c. Scalar and vector fields use different operators altogether. The universal language is therefore the shared null *characteristic cone*, not a hidden classical trajectory assigned to every particle and interaction channel.[4]

[3] For an operator whose symbol is $q(\omega, \mathbf{k})I + L$, with L independent of momentum, the characteristic set is controlled at principal order by $q = \omega^2 - |\mathbf{k}|^2$. The conclusion would require re-audit if a proposed channel supplied additional derivatives of the same or higher order.

[4] For the Dirac Hamiltonian, $\dot{x}_i = \alpha_i$ and each α_i has eigenvalues ± 1, but $[\alpha_i, \alpha_j] \neq 0$ for $i \neq j$. Componentwise spectral support therefore does not define a joint classical velocity vector. Massive wave-packet drift instead follows the dispersion relation and has magnitude $|\mathbf{p}|/E < 1$.

3 How mass appears when null directions disagree

ESTABLISHED PHYSICS Two future-directed null momenta can add to a timelike momentum. Draw two lightlike arrows from the same origin (Figure 2). If they point along the same ray, their sum remains lightlike. If they open apart, their sum moves inside the light cone and acquires invariant mass. This is the mirrored box of light from the opening section, drawn as momentum arrows: each flash is massless, but the pair is not.

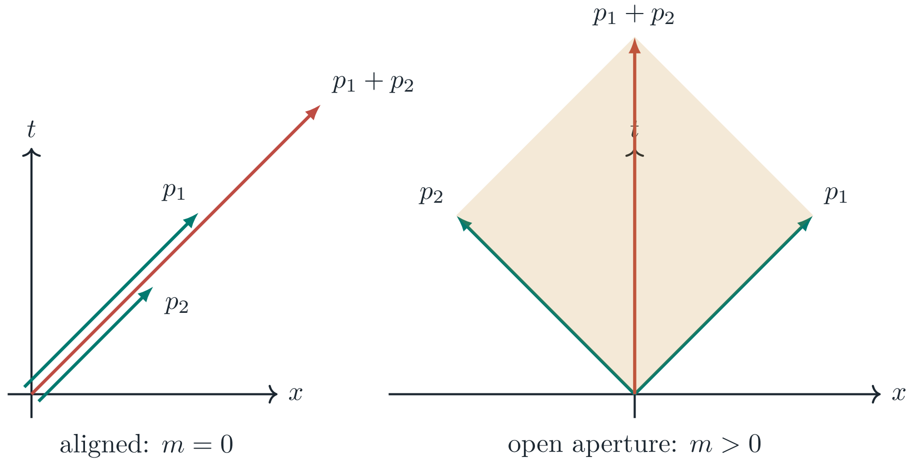


Figure 2: Alignment preserves nullness. Directional disagreement creates a timelike sum. The shaded area is a visual proxy for the wedge invariant. The aligned arrows are drawn slightly offset so both remain visible; they lie on the same ray.

The finite theorem uses a 2×2 positive momentum matrix

$$P = \sum_i \psi_i \psi_i^\dagger.$$

Each spinor ψ_i supplies one rank-one, hence massless, contribution. The determinant is

$$\det P = \sum_{i<j} |\psi_i \wedge \psi_j|^2.$$

Every term is nonnegative. The determinant vanishes exactly when all occupied spinors are collinear. In the standard momentum-bispinor normalization, $\det P = m^2$.[5]

checked finite result **In the finite model, mass squared is exactly the total pairwise failure of the occupied null directions to align.**

This is mathematically clean, but it is initially a kinematic statement. Any timelike momentum can be decomposed into null momenta, so the identity alone does not explain why a particle has one mass rather than another. A dynamical origin requires an operator, an action, a stable sector, and ultimately a continuum limit. Those are separate layers of the program.

[5] The identity is the 2×2 Cauchy–Binet formula: $\det(\Psi\Psi^\dagger) = \sum_{i<j} |\det(\psi_i, \psi_j)|^2$. Identifying the determinant with Minkowski norm uses the Pauli-matrix map $p_\mu \mapsto p_\mu \sigma^\mu$ and a fixed metric/sign convention.

4 Time in steps: the checkerboard picture

In one spatial dimension, a null step goes either right or left. A massive Dirac amplitude can be represented as a sum over zigzag histories (Figure 3). The everyday analogue is a sailboat tacking upwind: on every leg the boat moves at full sailing speed, yet its net progress up the channel is slower, set by how often it comes about. Each corner changes the propagation channel and contributes a factor proportional to the mass:

$$A(h) = (\mathrm{i}\epsilon m)^{\#\mathrm{corners}(h)}.$$

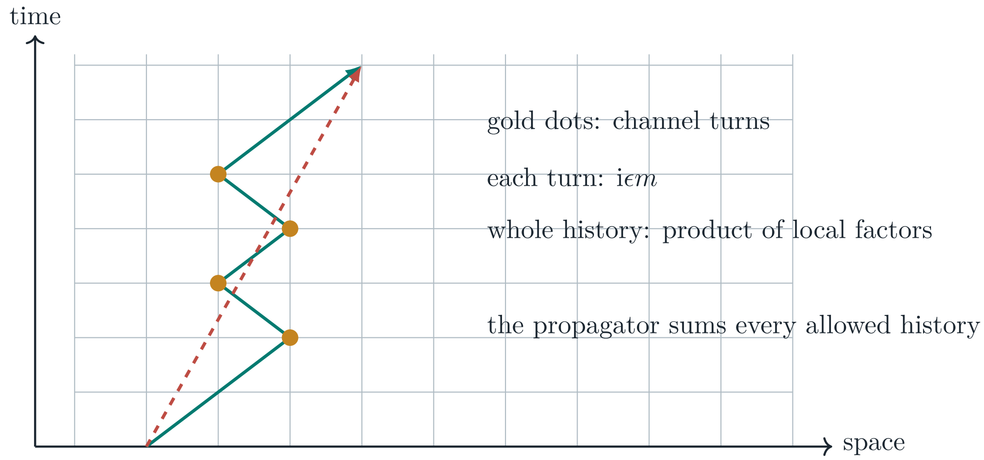


Figure 3: A finite checkerboard history. Mass enters at turns, while every segment remains null.

At $m = 0$, every history containing a corner has zero amplitude; only straight null histories survive. At nonzero m, turned histories participate and interfere. CHECKED FINITE RESULT The project has checked the exact finite sum and its discrete Dirac recursion using exact fraction arithmetic, so no rounding or numerical approximation enters that finite statement.[6]

The grid should be read as a regulator and a proof laboratory. A fixed finite direction set selects a frame and cannot by itself be exactly Lorentz invariant. The program does not infer that physical spacetime is literally a visible crystal. The continuum and ensemble limits must erase the regulator's preferred structure in the observables that survive.[7]

The first $3 + 1$ lift

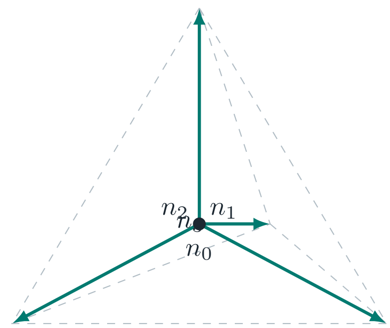
In three spatial dimensions, four directions pointing toward the vertices of a regular tetrahedron form the smallest isotropic set of this kind. If c_i counts the number of steps in direction i, the finite endpoint obeys

$$X(c)^2 = \frac{8}{3}\sum_{i<j} c_i c_j.$$

A history using only one direction is null. Mixing any two occupied directions creates a positive timelike interval.

[6] Formally, the computation runs over the Gaussian rationals: complex numbers whose real and imaginary parts are ordinary fractions. Every quantity in the finite statement is represented exactly.

[7] At finite spacing the symmetry is only the discrete symmetry of the direction set. Recovering Lorentz symmetry requires a scaling limit in which the propagator, not merely a fitted dispersion curve, converges with controlled error to a Lorentz-covariant continuum kernel.



$n_i \cdot n_i = 1$
$n_i \cdot n_j = -1/3$ for $i \neq j$
four null directions
one isotropic spatial frame

CHECKED FINITE RESULT There is a sharp normalization lesson. Equal averaging over these four unit directions produces $\frac{1}{3}$ of the identity on spatial vectors. The simplest spin-projector recursion therefore cannot simultaneously have unit microscopic step speed, that normalized average, and unit emergent wave speed. This is a constraint on one discretization, not a no-go against $3+1$ quantum walks.[8]

5 The four channels

The finite carrier has schematic form

$$D = \sum_e c(\alpha_e)\nabla_e + \Gamma\phi.$$

Here $c(\alpha_e)$ represents a null direction, ∇_e transports a state, and $\Gamma\phi$ changes an internal or chiral channel. Squaring D separates four structural effects.

Channel	Visual question	Finite mathematical content
Aperture	Do occupied null directions open apart?	Positive Gram/wedge terms; rank one becomes rank two; invariant mass appears.
Closure	Does transport around a loop return unchanged?	Commutators and holonomy; a closed path can return with a nontrivial phase or rotation.
Turn	Does propagation switch between null channels?	Chiral conversion and checkerboard corners; mass weights the switch.
Solder	Do internal frames and spacetime directions agree?	Coframe mismatch, torsion/nonmetricity-shaped terms, and a candidate geometry channel.

5.1 Aperture: opening a timelike interior

"Aperture" is the opening between occupied null rays. It is not a literal hole in space. Closed scissors present a single blade-line; opened scissors enclose a wedge of area between the blades. One ray has zero wedge area with itself. Two nonparallel rays span an area, and the squared area is the mass contribution. Aperture is the kinematic channel and the cleanest part of the framework.

[8]For tetrahedral unit vectors, $\sum_{i=0}^{3} n_i n_i^T = (4/3)I_3$, so equal averaging gives $I_3/3$. In the minimal projector ansatz this factor rescales the spatial principal symbol. An explicit direction register, different weights, or a more elaborate unitary walk can change the ansatz; the finite theorem forbids only the three requested normalizations within the simple recursion.

5.2 Closure: remembering a loop

Walk a large closed loop on the curved Earth carrying an arrow you never deliberately rotate: start on the equator pointing north, walk to the north pole, turn and walk back down to a different point on the equator, then walk home along it. The arrow comes back rotated, although you never twisted it; the loop itself did. Closure is the same experiment on the grid. Place an internal arrow at one grid point and transport it around a closed loop (Figure 4). The endpoint is the starting point, but the arrow may return rotated or with a phase. That residual transformation is the holonomy.[9]

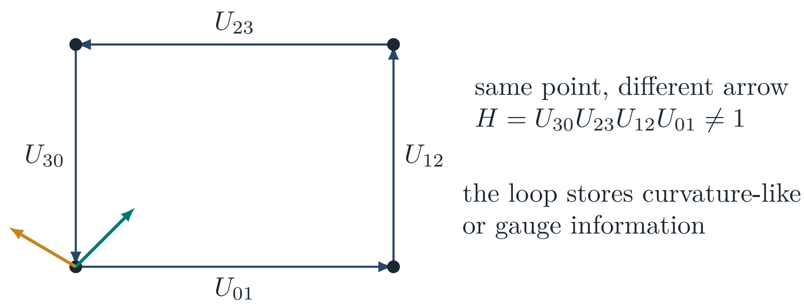


Figure 4: Closure is not the geometric closing of the square; it is the failure of transported internal data to return unchanged.

For complex $U(1)$ edge phases, an open path transforms only at its endpoints. A closed-loop product is gauge invariant. CHECKED FINITE RESULT The finite model contains an exact four-edge example with holonomy $\mathrm{i} \neq 1$ that stays i under a nonidentity gauge change.

CHECKED FINITE RESULT The nonabelian upgrade is now also checked. For matrix-valued transport, an open history transforms only at its endpoints; a closed loop is conjugated by the gauge at its basepoint rather than left invariant; and the loop's matrix trace survives exactly. An explicit rational 2×2 example with two noncommuting shear transports realizes all three statements: a nonidentity gauge change visibly moves the holonomy matrix while its trace stays fixed at 3. What remains open is connecting this history holonomy to the carrier's closure block, not the holonomy algebra itself.

5.3 Turn: converting one null channel into another

For a massless Weyl fermion, left- and right-handed propagation decouple. A Dirac mass couples them. In the checkerboard picture, the coupling is visible as a turn; in operator language, it is the off-diagonal conversion between two null chiral sectors. The path picture and the matrix picture are two views of the same local event. In the sailboat language of Section 4, turn is the tack: the mass sets how often the boat comes about.

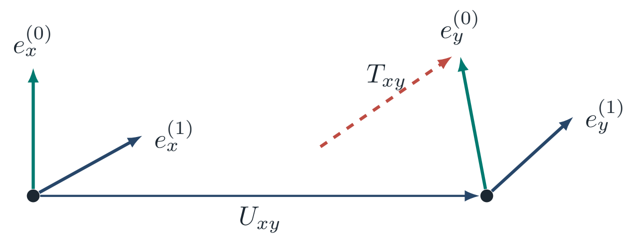
5.4 Soldering: tying transport to spacetime

Internal vectors do not become spacetime directions by naming them so. A printed map is useful only once its compass rose is aligned with north on the actual ground, and the alignment must be rechecked from place to place. A *coframe* is the mathematical map that performs this identification. At neighboring

[9]For an ordered path $e_1 \cdots e_n$, the transport is the ordered product $U_{e_n} \cdots U_{e_1}$. Under vertex gauge transformations, open-path transport changes by endpoint factors; for $U(1)$ a closed product is invariant, while for a nonabelian group it is conjugated and only conjugacy-class data such as a matrix trace is gauge invariant.

sites x and y, let e_x and e_y be coframes and U_{xy} the transport between their internal frames (Figure 5). Their mismatch is

$$T_{xy} = e_y - U_{xy} e_x.$$



transport the frame at x; compare it with the frame already at y

Figure 5: Soldering is the comparison between an internally transported frame and the local frame that defines spacetime directions.

CHECKED FINITE RESULT The finite matrix-coframe model now proves that invertible frame changes preserve coframe nondegeneracy, Lorentz frame changes preserve the induced metric $e^T \eta e$, determinant-one changes preserve volume, and the quadratic defect action $\mathrm{tr}(T^T \eta T)$ is frame invariant. An exact rational boost provides a nonzero witness.[10] This is meaningful finite geometry, but it is not yet general relativity: there is no continuum tetrad field, Einstein equation, or graviton in the result.

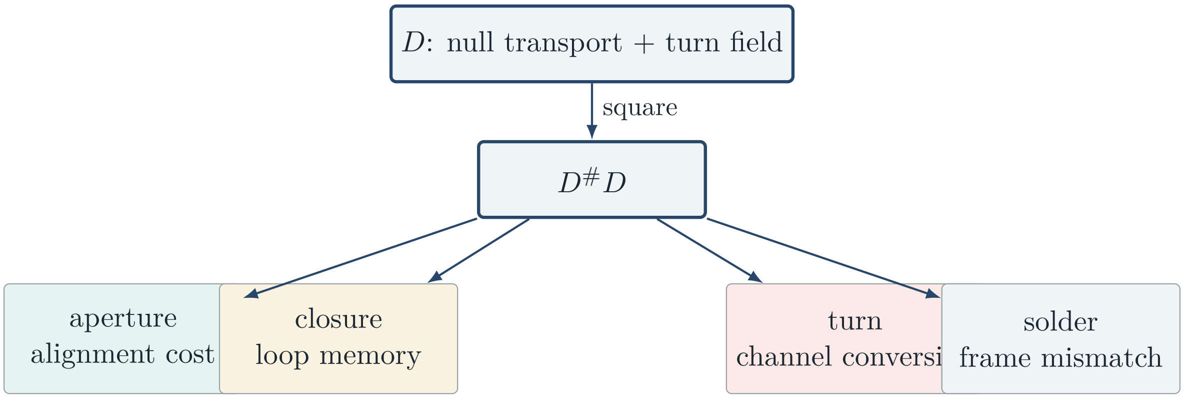
6 Why these four belong together

The unifying object is not the word “mass.” It is the square of one finite Dirac-like carrier. Schematically,

$$D^{\#} D = \underbrace{Q_A}_{\text{aperture}} + \underbrace{Q_C}_{\text{closure}} + \underbrace{Q_T}_{\text{turn}} + \underbrace{E}_{\text{soldering}} .$$

The four terms are distinguished by how transport, commutators, grading, and frame variation enter the square. CHECKED FINITE RESULT In one explicit rational carrier, the four blocks are linearly independent and their coefficients can be recovered from matrix entries. That makes the displayed decomposition rigid for that witness. It does not prove that every reasonable carrier has a unique four-channel decomposition; an abstract non-uniqueness theorem shows that an additional physical principle is needed.

[10] Here a coframe is represented by an invertible matrix e, the induced metric is $g = e^T \eta e$, and the edge defect is $T_{xy} = e_y - U_{xy} e_x$. The finite covariance theorem is algebraic. It does not supply differentiability, a continuum connection, or a field equation.



This is the framework's deepest simplification: instead of proposing four unrelated stories for mass, ask whether four familiar physical mechanisms are the four graded faces of one operator square. The finite algebra says that the question is coherent. Nature has not yet answered yes.

7 What the picture says about particles

The slogan "all mass comes from massless primitives" has different reach for different species.[11]

Object	Null-edge picture	Boundary
Massless fermion	One Weyl propagation channel; one null momentum direction.	Standard chiral kinematics.
Massive fermion	Two null chiral channels coupled by a mass term; zigzag drift.	Finite algebra checked; literal trajectory remains interpretation.
Photon	Massless vector with only transverse polarizations.	The velocity-operator story for Dirac fermions does not transfer verbatim.
Massive W/Z	The longitudinal polarization appears together with mass; a would-be scalar mode is reorganized into the vector field.	Degree-of-freedom accounting is standard; the null-edge dynamics is not derived.
Neutrino	Dirac mass links an independent partner; Majorana mass links a state to its charge conjugate.	Which option nature uses remains unknown.
Antiparticle	CPT-related orientation of the same relativistic structure.	No explanation of the cosmic matter asymmetry follows.
Composite matter	Constituents and binding fields contribute to total rest energy.	A composite object is not claimed to move microscopically at c.
Higgs scalar	No Weyl zigzag of its own.	The scalar self-mass is an explicit exception to the universal fermionic picture.

The Higgs row matters. The framework can describe how a scalar expectation reorganizes fermion and

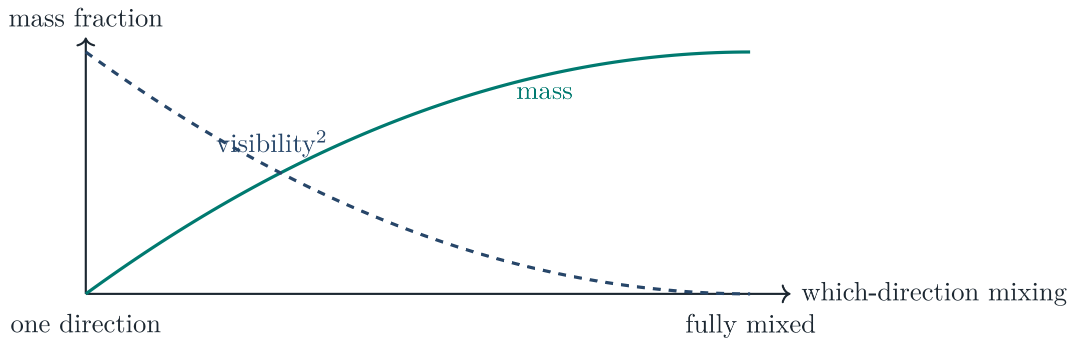
[11] The broadest theorem available here is kinematic: a future-timelike momentum can be decomposed into future-null momenta, and the finite determinant measures their nonalignment. A dynamical explanation of a measured rest mass must additionally select the state and its coupling. The fermionic checkerboard supplies such a mass-weighted turn mechanism in a finite model, but it does not by itself cover scalar self-mass, confinement energy, or the observed Yukawa spectrum.

vector degrees of freedom, but it does not derive the Higgs boson's own mass from a pair of null chiral channels. A universal slogan that hides this exception would be shorter and less scientific.

8 Mass as lost directional simplicity

The same 2×2 determinant has an information-theoretic reading. Here "visibility" means what it means in the double-slit experiment: the contrast of interference fringes, which fades as which-path — here, which-direction — information becomes available. A state supported on one perfectly known direction is rank one and has zero determinant. Mixing distinguishable directions raises the determinant. In the simplest two-direction model,

$$\text{mass fraction} + \text{visibility}^2 = 1.$$



This does not mean that ordinary environmental noise literally creates a particle's rest mass. It means that the finite invariant used as the mass proxy is also a measure of retained directional distinguishability. Indeed, the model contains a signed closure operation that can lower the determinant; closure is structured transport, not generic decoherence.

The information view is useful because it unifies several descriptions: rank defect, linear entropy, loss of visibility, total-variation distinguishability, and compression cost all vanish on a single pure direction. They are related faces of one invariant, not independent confirmations of the physical theory.

9 What the machine has checked

What is Lean? Lean is a proof assistant: a computer program in which definitions, theorems, and proofs are written in a formal language and checked, step by logical step, by a small trusted core called the kernel. A statement the kernel accepts cannot hide a gap, a sign error, or an unstated assumption, which is stronger than "we did the algebra carefully." What the kernel cannot check is whether the formal statement means what the surrounding prose says it means.

Lean verifies formal statements, not the intended interpretation. The project therefore treats proof and semantic audit as separate obligations. The current evidence ladder is:

Kernel-checked finite results Gram/wedge mass identity; massless iff aligned; exact 1 + 1 checkerboard sum; 3 + 1 tetrahedral endpoint law; one-step continuum error bound; subluminal massive drift; carrier four-block witness; finite binding; abelian and nonabelian closure holonomy; nondegenerate matrix-coframe covariance.

Established external physics Massive momenta as sums of null momenta; Dirac/Weyl chirality; Feynman checkerboard and quantum walks; gauge holonomy; coframes and Lorentz geometry; Higgs degree-of-freedom transfer.

Open physical identifications The finite carrier is nature's microscopic carrier; its four blocks become Standard Model kinetic/gauge/Yukawa/gravity terms; the regulator has the correct continuum and many-body limits; observed mass values follow.

Some particularly visual checked statements are worth listing plainly.

- One null direction gives zero mass invariant; two nonparallel directions give a strictly positive rational witness.
- Every primitive tetrahedral step is null, while a mixed two-direction endpoint is timelike.
- The basic tetrahedral projector average has an unavoidable factor 1/3; the normalization tension is a theorem, not a plotting artifact.
- A finite massive shell has group drift strictly below c, including an exact 3–4–5 witness with speed squared 9/25.
- A closed finite $U(1)$ loop can have nontrivial, exactly gauge-invariant holonomy.
- For noncommuting matrix transport, a closed loop's holonomy is conjugated by the basepoint gauge — an explicit rational gauge change moves the holonomy matrix — while its trace is exactly unchanged.
- A rational nonidentity Lorentz boost preserves the induced metric and a nonzero matrix-coframe defect action.

10 What could prove the picture wrong

A useful speculative framework should become easier to kill as it becomes more specific. The main failure modes are visible now.

1. **No continuum recovery.** If the finite walk does not converge to the Dirac/Weyl propagator with controlled errors, the grid is only a toy.
2. **Wrong characteristic cone.** If channel terms enter at principal order and alter the observed null cone, the universal-speed reading fails.

3. **Wrong field dictionary.** If aperture, closure, turn, and soldering do not refine to kinetic, gauge, scalar, and geometric operators, the physical names must be replaced by neutral block labels.
4. **No many-body locality.** Single-particle quantum walks do not automatically produce a local quantum field theory. Known higher-dimensional QCA obstructions must be confronted rather than bypassed.
5. **No mass values.** Structural mass identities do not predict the electron, quark, or neutrino spectrum. A successful theory needs a rigid generation/Yukawa mechanism, not inserted constants.
6. **No gravitational limit.** A finite coframe defect is not yet an Einstein or teleparallel field equation. Failure of the soldering channel to produce a legitimate continuum action would retire the gravity reading.
7. **Scalar exception remains unexplained.** The Higgs self-mass must be derived by another layer or retained openly as an input.

A theorem about a finite matrix can be completely correct while its proposed physical interpretation is wrong. The program is designed so that those two verdicts can be stated independently.

11 The research program ahead

The next conceptual milestones are concrete.

1. Sum the now-exact ordered $3+1$ spin-projector amplitudes into a full endpoint kernel; bend attenuation and orientation-dependent phase are landed.
2. Build an exactly norm-preserving tetrahedral walk with an explicit direction register and massive channel conversion.
3. Prove a many-step continuum theorem rather than only a one-step error estimate.
4. Instantiate the common principal-symbol theorem separately for scalar, Dirac, and gauge-fixed vector fields.
5. Connect the now-checked nonabelian history holonomy to the carrier's closure block on each history.
6. Relate the nondegenerate finite coframe to the actual soldering term in the carrier square and then to a continuum variational equation.
7. Derive, or sharply rule out, a mechanism for three generations and dimensionless mass ratios.

The goal is not to make the grid increasingly ornate. It is to discover which parts of the visual picture survive every change of regulator and become continuum invariants. Those survivors, if any, are candidates for physics.

12 Conclusion

The familiar statement "massive things move slower than light" describes observed drift. It need not describe the most elementary support of a quantum amplitude. A finite null-step model offers a different decomposition:

Null primitives Every elementary propagation link lies on the light cone.

↓

Structured disagreement Directions open, loops remember, channels turn, and frames fail to match.

↓

Massive emergence The combined state has timelike momentum, subluminal drift, and a definite rest energy.

This is a compelling way to see mass: not as a primitive substance and not as a light ray that has literally slowed down, but as the invariant cost of making many null processes coexist as one stable object. The finite mathematics now supports substantial parts of that sentence. Whether nature uses the same architecture is the question the remaining theorems and kill tests are meant to answer.

A Selected formal anchors

The public-facing discussion above is grounded in the following Lean modules. These are finite draft modules with build-enforced assumption-footprint checks; their physical interpretation remains subject to the boundaries stated in the text. A technical companion manuscript (*Null-Edge P1: The Origin of Mass*, draft v3) states the same results with full definitions, conventions, and provenance. Module paths in the table are relative to `PhysicsSM/Draft/NullEdge/` in the project repository (`https://github.com/Pandaemonium/StandardModel`); readers with a Lean 4 toolchain can rebuild and re-verify every listed result.

`Module`	Payload used here
`GateI1/Core.lean`	Gram/wedge mass identity and massless boundary
`ExactCheckerboardPathSum.lean`	Exact finite corner-weighted history sum
`TetrahedralNullHistory.lean`	$3 + 1$ endpoint law and $1/3$ frame factor
`TetrahedralSpinProjectorPath.lean`	Ordered bends and orientation-sensitive spin phase
`LowerOrderChannelCausality.lean`	Shared null principal cone and lower-order gate
`DiracVelocityOperator.lean`	Componentwise Dirac velocity support
`Carrier/SubluminalBound.lean`	Massive group-velocity bound
`FourChannelRigidityCapstone.lean`	Concrete four-block coefficient rigidity
`HistoryLocalFourChannelAction.lean`	Local turn action and corner phase
`U1HistoryClosureHolonomy.lean`	Gauge-invariant finite closure phase
`NonabelianHistoryClosureHolonomy.lean`	Nonabelian closure: endpoint covariance, basepoint conjugation, gauge-invariant trace
`NondegenerateSolderingGeometry.lean`	Matrix coframe, metric, volume, and defect action
`Carrier/DerivedInteraction.lean`	Closure-derived finite binding and wrong-plane control

B Glossary

Amplitude. The complex number quantum mechanics attaches to each way an event can happen. Amplitudes for indistinguishable alternatives add before being squared into probabilities, which is why histories interfere.

Null / lightlike. On the light cone: a step or momentum that covers exactly one unit of distance per unit of time (in units where $c = 1$), so its spacetime interval is zero.

Timelike. Inside the light cone: slower than light. The average motion of every massive object is timelike.

Invariant mass. The mass every observer agrees on, defined by $m^2 = E^2 - |\mathbf{p}|^2$. Zero for a single flash of light; positive for a system of misaligned flashes.

Chirality. The two independent propagation channels ("left-handed" and "right-handed") of a relativistic fermion. Massless fermions keep the channels separate; a mass term couples them.

Weyl / Dirac fermion. A Weyl fermion is a single massless chiral channel; a Dirac fermion is two chiral channels coupled by a mass.

Propagator. The table of amplitudes for getting from one place and state to another in a given number of ticks.

Regulator. A deliberately simplified setting — here the finite grid — that makes every quantity exactly computable. Results are trusted as physics only if they survive removing the regulator.

Holonomy. The net rotation or phase an internal arrow acquires when transported around a closed loop.

Coframe / soldering. The map that identifies internal directions with actual spacetime directions, and the requirement that this identification be consistent from place to place.

Principal part / front speed. The highest-derivative part of a wave equation, which alone controls the fastest possible wavefront.

Group velocity. The speed of the peak of a wave packet; the observed drift of a massive particle, always below c.

Proof assistant / kernel. See the box in Section 9.

C Further reading

The first four entries are accessible starting points for a general reader; the rest are technical anchors for the specific mechanisms discussed above.

References

[1] R. P. Feynman, *QED: The Strange Theory of Light and Matter*, Princeton University Press, 1985.

[2] F. Wilczek, *The Lightness of Being: Mass, Ether, and the Unification of Forces*, Basic Books, 2008.

[3] F. Wilczek, *Mass without mass I: most of matter*, Physics Today 52(11) (1999) 11; *Mass without mass II: the medium is the mass-age*, Physics Today 53(1) (2000) 13.

[4] R. Penrose, *The Road to Reality*, Jonathan Cape, 2004.

[5] R. P. Feynman and A. R. Hibbs, *Quantum Mechanics and Path Integrals*, McGraw–Hill, 1965.

[6] T. Jacobson and L. S. Schulman, *Quantum stochastics: the passage from a relativistic to a non-relativistic path integral*, J. Phys. A 17 (1984) 375.

[7] B. Z. Foster and T. Jacobson, *Spin on a 4D Feynman Checkerboard*, arXiv:1610.01142.

[8] G. M. D'Ariano, N. Mosco, P. Perinotti, and A. Tosini, *Path-sum solution of the Weyl Quantum Walk in 3+1 dimensions*, Phil. Trans. R. Soc. A 375 (2017) 20160394, arXiv:1705.08552.

[9] U. Nzongani et al., *Dirac quantum walk on tetrahedra*, arXiv:2404.09840.

[10] N. Arkani-Hamed, T.-C. Huang, and Y.-t. Huang, *Scattering Amplitudes For All Masses and Spins*, JHEP 11 (2021) 070, arXiv:1709.04891.

[11] J. C. Baez and D. K. Wise, *Teleparallel gravity as a higher gauge theory*, Commun. Math. Phys. 333 (2015) 153–186, arXiv:1204.4339.

[12] L. Mlodinow and T. A. Brun, *Quantum field theory from a quantum cellular automaton in one spatial dimension and a no-go theorem in higher dimensions*, Phys. Rev. A 102 (2020) 042211, arXiv:2006.08927.